GUJARAT TECHNOLOGICAL UNIVERSITY

Chandkheda, Ahmedabad

GOVERNMENT ENGINEERING COLLEGE

BHARUCH

A report on

"Drone Cam: Drone with Adaptive Mounting System for Cinematography"

Under subject of

DESIGN ENGINEERING – 2B

B.E. III, Semester - VI

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

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ACADEMIC YEAR: 2025-26

GOVERNMENT ENGINEERING COLLEGE BHARUCH

DEPARTMENT OF
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CERTIFICATE

This is to certify that the Report entitled **“Drone Cam: Drone with Adaptive mounting system for cinematography”** has been carried out by Rachit V. Kulkarni (230140111038), Mallah Tejas Rajkumar (230140111020), Vardhan R. Patel (240143111010), Bhavya B. Patel (240143111008) following students under my guidance in fulfilment of the term work of Bachelor **of Engineering** in **ELECTRONICS AND COMMUNICATION 4th Semester** of Gujarat Technological University, Ahmedabad during the academic year 2025-26.

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We would like to extend our sincere thanks to our internal guide Prof. Devendra R. Patel for their useful guidance for our project. He guided us very much for project and report.

This movement asks for a token of gratitude to them for giving us this opportunity to undertake a project firm. Our heart full thanks to you sir.

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ABSTRACT

The project, titled Drone Cam: Adaptive Mounting System for Cinematography, aims to address the growing demands of advanced drone technology in filmmaking. It facilitates easy adjustments to camera angles and enhances aerial filming capabilities.

To achieve these goals, the project utilized a blend of mechanical design, aerodynamic optimization, and camera stabilization techniques. The approach included developing a prototype with adjustable mounts, which were tested under various environmental conditions to evaluate stability and image quality.

The adaptive mounting system offers greater flexibility for filming while reducing the need for manual adjustments, making it ideal for dynamic cinematography and aerial applications.

This technology holds the potential to enhance film production, surveillance, and precision camera work across various industries.

The adaptive drone system represents an innovative solution for cinematography, potentially impacting both drone technology and media production.

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Chapter 1

INTRODUCTION

In today's world, drones are transforming how we capture visuals from the sky. The DroneCam project focuses on creating a drone with an adaptive mounting system designed specifically for cinematography. This system allows the camera to easily adjust its position and angle, giving filmmakers smoother and more dynamic shots.

Traditional camera setups often have limitations — they can't reach certain angles and require complex setups. Drones solve this problem by providing flexibility, mobility, and precision in capturing footage from unique perspectives. The DroneCam enhances this experience even further with a design that improves stability, control, and battery efficiency, ensuring high-quality results.

With advancements in stabilization systems and adaptive mounts, aerial cinematography has become more creative and efficient. Drones are not only changing the film industry but are also finding uses in areas like real estate, agriculture, surveillance, and environmental monitoring. The DroneCam project aims to make this technology more accessible and practical for both professionals and enthusiasts, encouraging innovation in visual media.

This project focuses on the development of a quadcopter drone system designed for stable and controlled flight using a custom-built control setup. The system integrates brushless DC motors, electronic speed controllers, and a manually designed radio communication system using Arduino Nano and nRF24L01 modules.

Unlike conventional drones that rely on commercial flight controllers and transmitters, this project emphasizes building a low-cost and customizable solution. The drone uses an MPU6050 sensor for stabilization, enabling better orientation control during flight.

The aim of this project is not only to understand drone mechanics but also to explore real-time embedded system integration, wireless communication, and control systems.

Chapter 2

LITERATURE REVIEW

Drone technology has rapidly evolved over the past decade. Initially, drones were mostly used for recreation and basic photography, but now they play a major role in transportation, delivery, research, and filmmaking.

Our research showed that although there are many types of drones available, none offer a truly adaptive camera mounting system that meets the needs of cinematographers. Most existing drones have fixed mounts, which limit camera movement and creativity. Moreover, low-cost drones often lack stability and obstacle avoidance, which can lead to poor footage or crashes.

The DroneCam project aims to fill this gap by designing a drone that is both functional and educational. It is meant for cinematographers, who need customizable mounting options for their cameras, and also for students and hobbyists, who can learn the basics of drone technology while exploring creative ways to capture visuals.

Through this approach, DroneCam combines practical learning with real-world applications, showing how engineering design can improve both usability and innovation in drone-based cinematography.

Chapter 3

EMPATHY MAPPING

Understanding real problems in society is honestly one of the toughest parts for us engineering students. Until now, most of our projects were based on imaginary ideas or technical assumptions, so we rarely got to see how things actually affect people's lives. This canvas activity really changed that. It was about stepping out of the classroom mindset and into the real world—talking to people, listening to their problems, and understanding their daily struggles. It wasn't a formal interview or a technical discussion; it was more of a casual, friendly conversation. The idea was to make people feel comfortable enough to open up and share what really matters to them. This whole process was called **Storyboarding Canvassing**.

The goal was to define a **user-centric problem**, and to do that, we first had to truly understand the user. During our brainstorming, we thought about many types of workers for our wireless monitoring concept. But then we realized we wanted to focus on those who are usually left behind by technology—people who make up a huge part of the workforce but rarely get access to modern solutions.

Next came identifying the **stakeholders**, the people directly or indirectly connected to the problem. After some discussion, we listed out everyone who plays a part or is affected by it in some way.

Then we moved to the **"Activities"** section, where we noted down all the steps we took to gather information and insights for our topic.

The last and most interesting part was the **"Storyboards"**—we called it the board of emotions. This helped us see the human side of the problem. We tried to understand how workers actually feel while working in their usual conditions. It made us realize that when we design something, it's not just about solving a technical issue—it's about understanding the emotions behind it too. We came up with several stories, especially related to the *four-hacksaw blade*, capturing both happy and sad moments that really touched us.

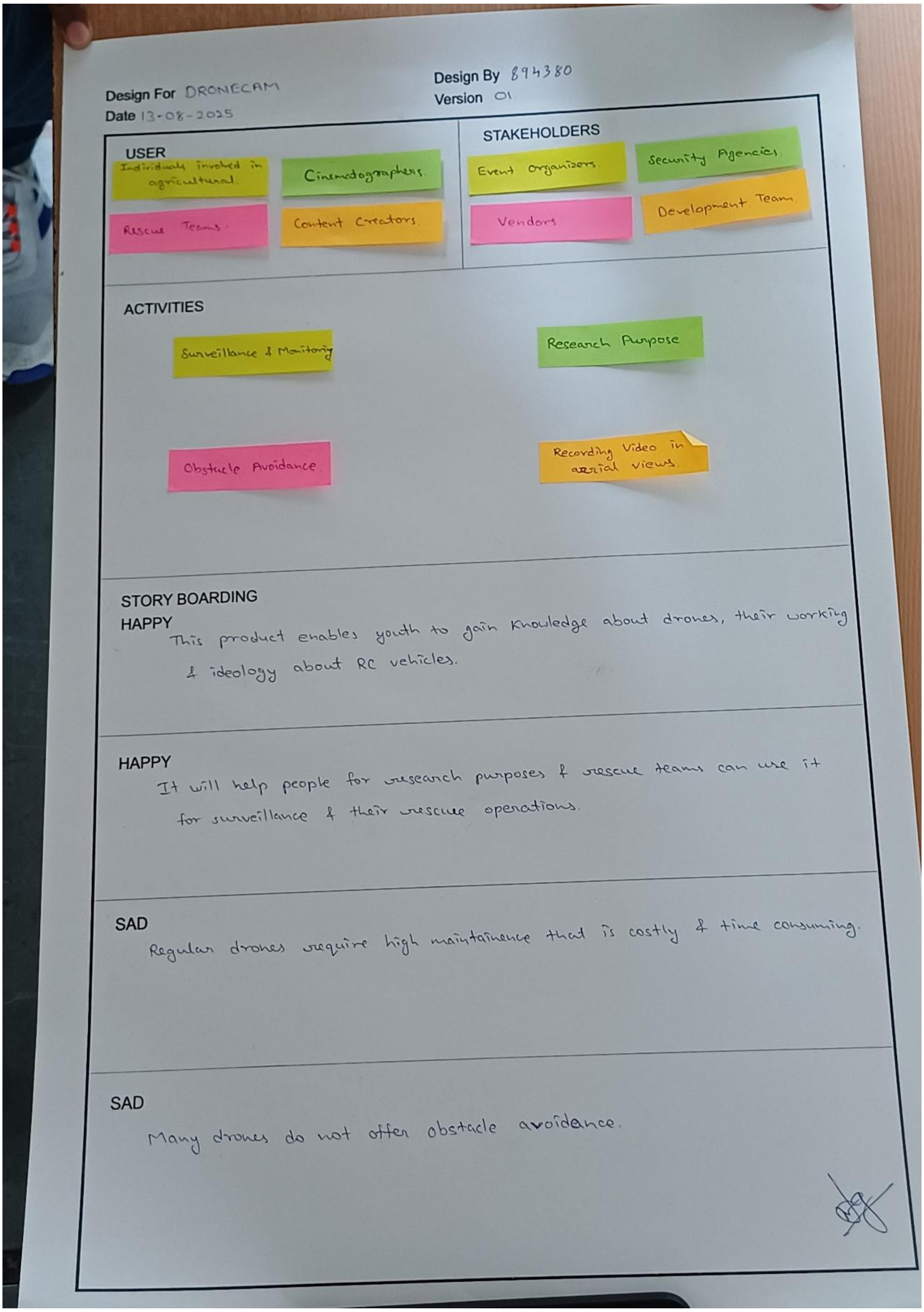

Figure 1 EMPATHY CANVAS

3.1 Storyboarding

HAPPY: This product helps young people understand drones — their working, design, and the basics of RC machines.

HAPPY: It can also support research and rescue operations. Teams can use it for aerial surveillance and locating people in disaster areas.

SAD: Regular drones need frequent maintenance. Repairs are costly and often take a lot of time.

SAD: Many low-cost drones lack obstacle avoidance, which causes crashes and damage.

3.2 User

Major users include farmers and agricultural professionals who use drones for crop monitoring and field analysis. Filmmakers and videographers rely on them for aerial shots, while rescue teams use them to find people in disaster zones. Content creators also use drones to produce creative aerial footage for online platforms.

3.3 Activities

Drones are used in surveillance, research, and monitoring across many industries like agriculture, security, and environmental studies. They're also common in media and event coverage for capturing aerial views. Modern drones now feature obstacle avoidance systems, making them safer and more efficient to operate.

3.4 Stakeholders

Key stakeholders include vendors who supply equipment, technicians who maintain and repair drones, and event organizers who use them for better coverage. Security agencies also use drones for patrolling, crowd monitoring, and public safety during large events.

Chapter 4

IDEATION

After understanding the users and their activities, the next step was to identify the problems faced by workers. To do that, we first discussed possible contexts, situations, and locations where these problems might occur. Only after exploring these real scenarios could we understand how different tasks are carried out and where improvements are needed.

Engineering, at its core, is about making people's lives easier through technology—reducing effort and improving efficiency. We looked into the challenges workers face under different conditions and then moved to the next part: finding possible solutions. This involved listing all potential ways technology, especially drones, could help overcome these issues.

Our next major task was to create the Ideation Canvas, where we described people's activities and the situations in which they experience problems. This canvas proved extremely useful—it helped us pinpoint the exact areas that needed attention and guided us toward a clearer project direction.

4.1 People

Different groups benefit from using drones in various ways. Rescue teams use them to access and inspect areas that are hard to reach during emergencies. Farmers and agricultural experts use drones to monitor crop health and soil conditions. Filmmakers rely on them for cinematic aerial shots, while hobbyists and enthusiasts fly and modify drones for fun and exploration. Content creators also depend on drones to produce engaging aerial visuals for their platforms.

4.2 Activities

Drones are used for a wide range of meaningful activities. The most common is aerial video capture, which provides stunning visuals for both media and personal projects. Researchers use drones to collect data from difficult environments, and security teams rely on them for surveillance and monitoring. Drones are also becoming a tool for education, inspiring students to explore technology hands-on. Modern drones now come with obstacle avoidance features, allowing safer and smarter navigation.

4.3 Situation / Context / Location

Drones are employed in many real-world environments. In commercial production, they're used to capture dynamic shots for films and advertisements. Broadcast crews use drones to cover live events from unique angles. Weddings, concerts, and sports events benefit from impressive aerial views. In agriculture, drones help survey large fields and provide data to improve crop management and productivity.

4.4 Props

Several key components make drone operation possible. The camera mount system ensures smooth and stable footage. The remote controller allows manual control, while the axis control board maintains flight stability. Propellers and BLDC motors with speed controllers provide thrust and control. All of these parts are housed within a lightweight yet durable frame, designed to balance performance and functionality.t

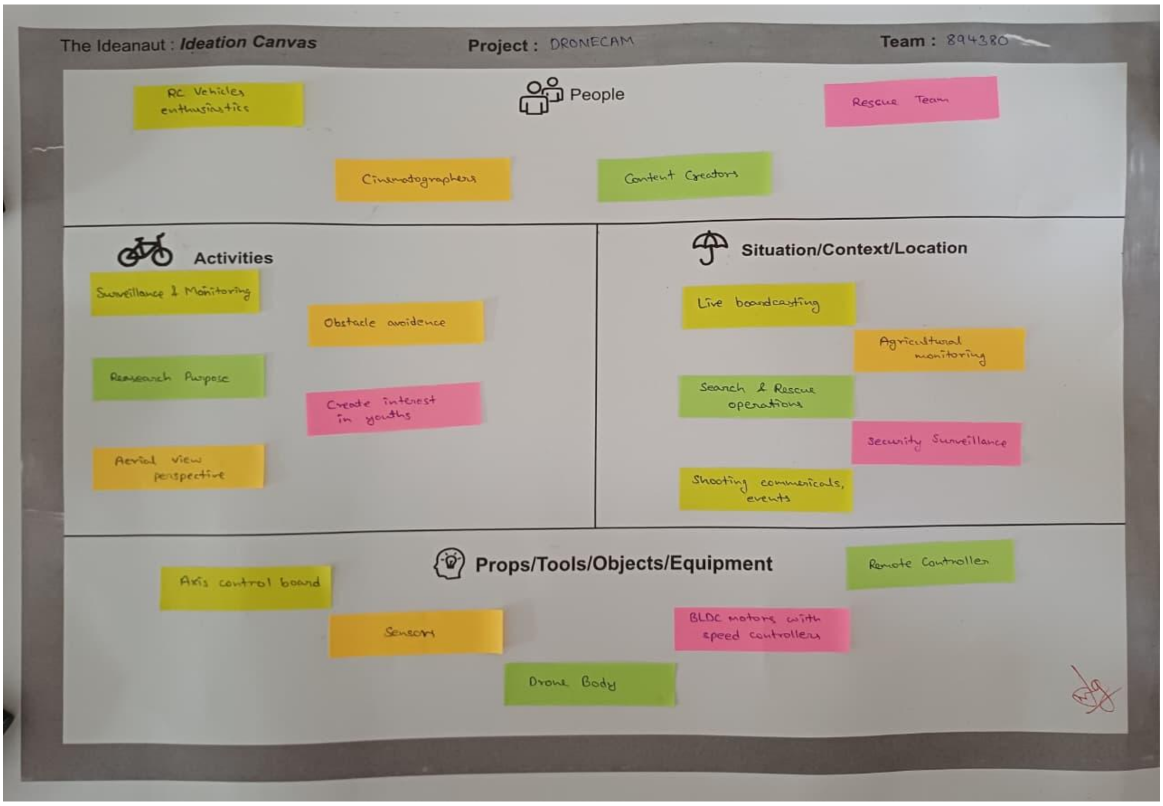

Figure 2 IDEATION CANVAS

Chapter 5

PRODUCT DEVELOPMENT CANVAS

Now we reached the core part of the design process — creating a product based on a key solution. A key solution, as we understood it, is one that directly addresses a key problem in a meaningful way. From the many possible solutions we brainstormed during the Ideation phase, we focused on the ones that would bring real, emotional value to the users.

This canvas highlights the solution development process — defining the purpose, identifying the user, listing the main features, explaining how the product functions, and noting the essential components involved.

5.1 Purpose

The drone serves several purposes: it helps in research and data analysis, supports search and rescue operations in difficult terrains, and sparks interest in drone technology among students and young learners.

5.2 Product Function

The drone can capture high-quality aerial videos while maintaining a stable flight. It is remotely operated, making it easy to control and suitable for various practical applications.

5.3 Product Feature

Its key features include detachable propeller arms for easy transport, a modular design that fits multiple uses, simple maintenance, and a replaceable camera mount for greater flexibility.

5.4 Components

The main components include a microcontroller for control logic, an axis control board for stability, a remote controller for user input, and BLDC motors with speed controllers for smooth and precise motion.

5.5 Customer Revalidation

The product especially appeals to young users who enjoy creativity and robotics. It’s also affordable, making it accessible for hobbyists, students, and beginners in drone technology.

5.6 Reject / Retain / Redesign

One limitation is the battery life, which affects the performance of the transmitter and receiver. The operating range also depends on the modules used. These aspects may require redesign or enhancement to improve efficiency.

5.7 Product Experience

Users appreciate its easy repair and upgrade options. The compact design ensures good portability, while its stable flight and mid-range capability provide a smooth and consistent flying experience.

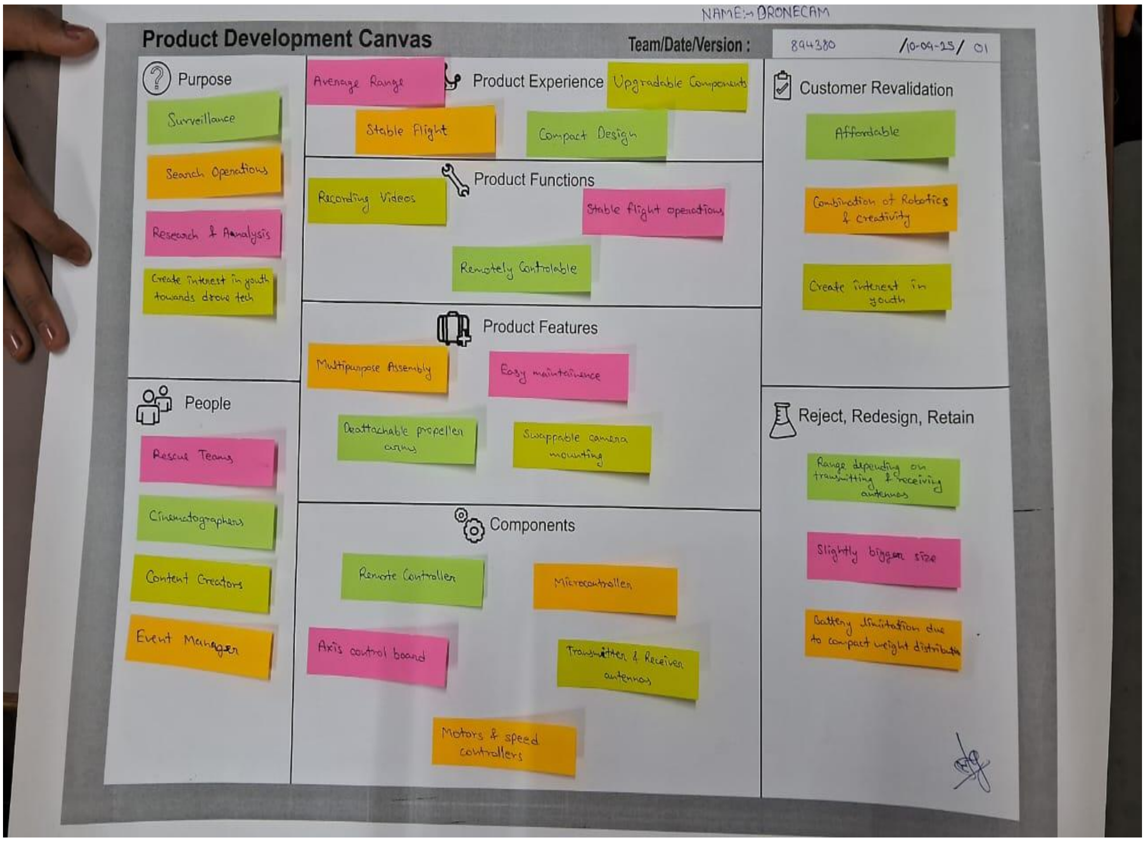

Figure 3 PRODUCT DEVELOPMENT CANVAS

Chapter 6

AEIOU CANVAS

(ACTIVITIES, ENVIRONMENT, INTERACTIONS, OBJECTS, USERS)

6.1 Activities

The development process involves simulating circuits using software tools, assembling the components on the drone frame, and testing the connection between the drone and its remote controller. Once assembled, test flights are carried out to check the drone's overall performance and functionality.

6.2 Environment

Drones are mainly operated in open outdoor spaces for safety and free movement. They're also used in research environments to collect data and in secured zones for surveillance, monitoring, or emergency response activities.

6.3 Interactions

The drone communicates wirelessly with its remote controller, allowing real-time control. Inside the drone, microcontrollers coordinate with other modules to ensure smooth and synchronized operation. The user interacts directly through the manual remote-control interface.

6.4 Objects

Key components include antennas and microcontrollers for transmission, an axis controller for flight stability, and a remote controller for manual operation. The drone is powered by Li-Po or Li-ion batteries and built on a lightweight but durable frame for better performance and endurance.

6.5 Users

Main users include farmers, who use drones for crop and field monitoring; rescue teams, who rely on them for search and recovery missions; and aerial photographers and hobbyists, who use drones for capturing visuals and exploring new perspectives.

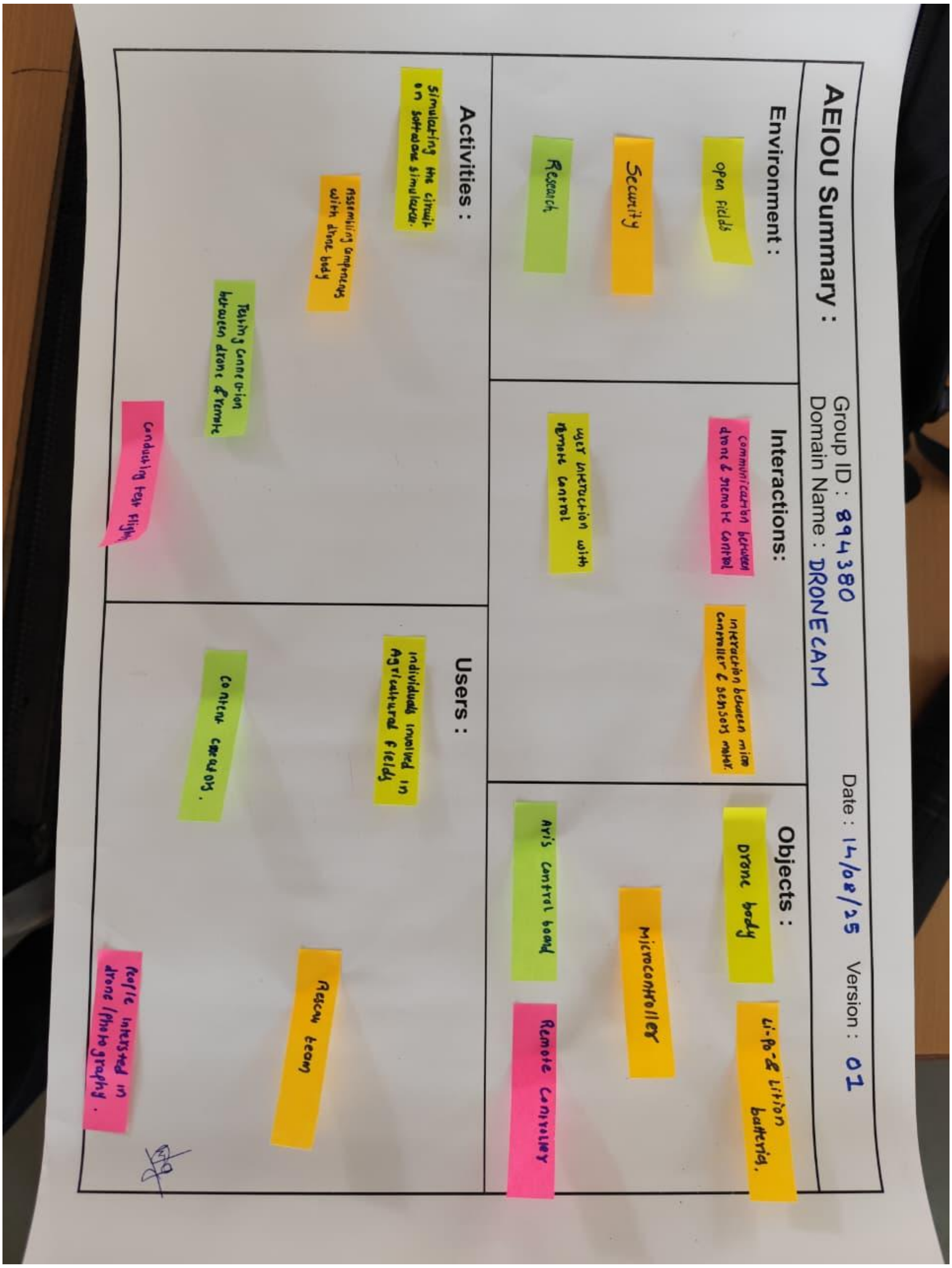

Figure 4 AEIOU CANVAS

Chapter 7

LNM Matrix

7.1 Design Specifications

The drone's design follows strict safety standards, with initial testing done through simulations. During development, designers carefully consider physical constraints and ensure the frame is sturdy enough to withstand minor crashes or rough landings.

7.2 Theories Involved

Several theoretical principles guide the design and working of the drone. Aerodynamics ensures efficient flight, while stability principles help maintain smooth motion. Axis control manages directional balance, and data transmission theories support communication between the drone and the controller.

7.3 Software / Tools / Skills Required

Development uses tools like Arduino IDE for programming and EasyEDA for circuit design and simulation. Key skills include circuit interfacing, component integration, and ensuring all electronic modules work together reliably.

7.4 Component Material / Strength Criteria

The drone frame is made from lightweight yet durable materials such as polycarbonate. It uses BLDC motors with speed controllers for stable propulsion and operates on Li-Po or Li-ion batteries for efficient power delivery and longer flight time.

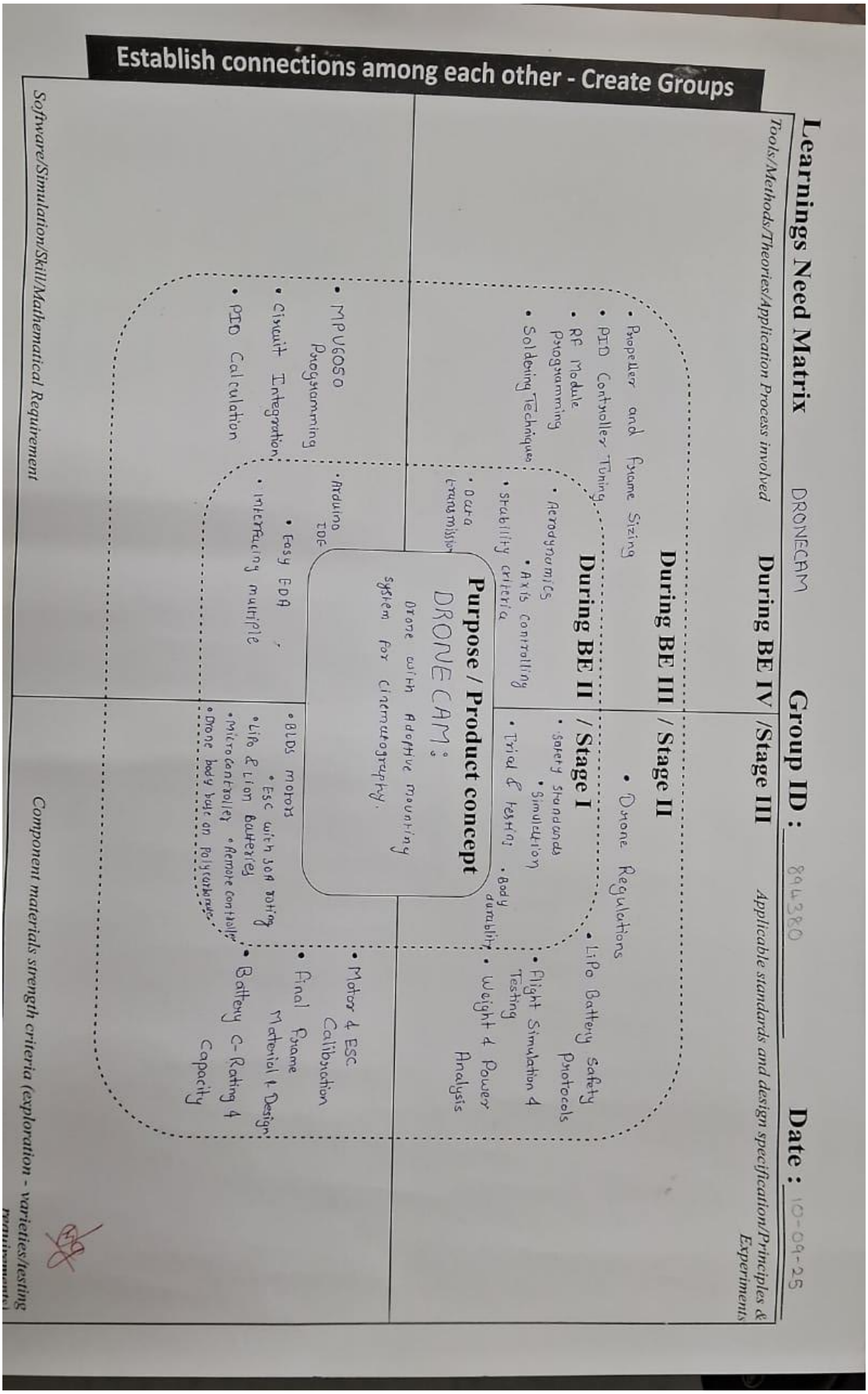

Figure 5 LNM MATRIX

Chapter 8

MIND MAPPING

8.1 Mind Map Flowchart

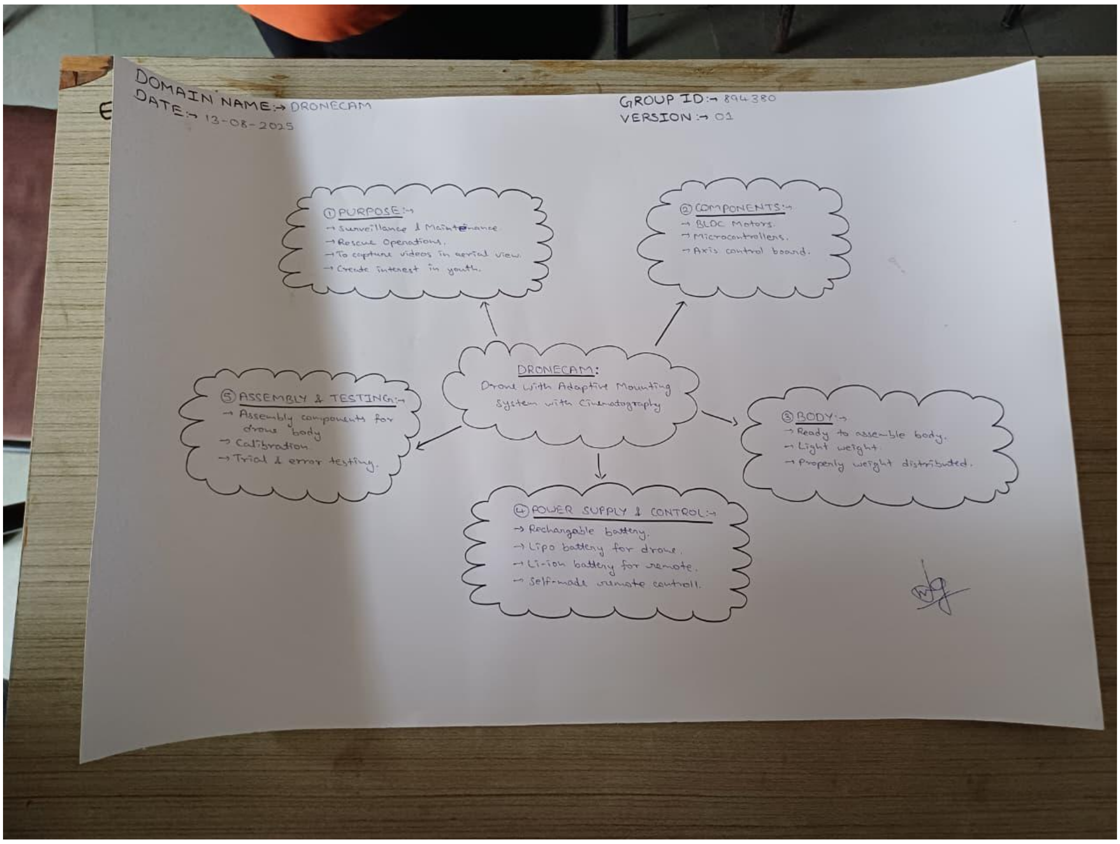

Figure 6 MIND MAP

8.2 Application

- Film & Video Production.
- Safety Security.
- Environmental Studies.
- Farming & Agriculture.
- Construction & Land Surveying.
- Disaster Response.

Chapter 9

PROTOTYPE

Drones are remote-controlled flying devices that have become popular in filmmaking, security monitoring, and scientific research, thanks to their ability to capture footage from different perspectives. For our project, we designed a drone equipped with an adjustable camera mount, allowing it to perform simple aerial maneuvers while staying stable for filming. The prototype was built using basic materials and components to demonstrate the core functions of a drone camera with a custom-built controller.

The drone is powered by a Lithium Polymer (Li-Po) battery, chosen for its lightweight design and high energy efficiency. The propellers are driven by motors controlled through Electronic Speed Controllers (ESCs), which regulate motor speed for precise control. The Arduino serves as the central control system, managing both movement and stability. This setup ensures smooth operation and easy handling, making it well-suited for our project's objectives. All components are connected using simple circuits, keeping the design straightforward while effectively showcasing the essential functions of a real drone camera.

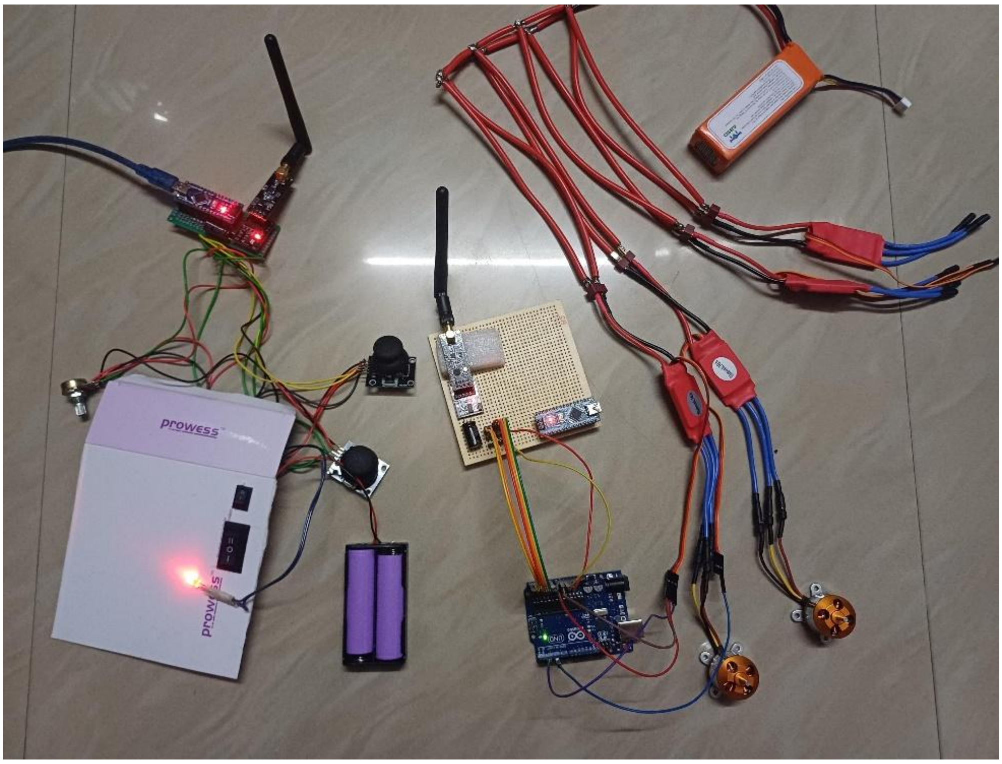

Figure 7: Prototype

The prototype has been developed using a TBS 500 quadcopter frame, which provides better structural support and balanced weight distribution compared to initial testing set-ups.

All four BLDC motors are mounted securely on the frame arms and connected to 30A ESCs. The central section houses the Arduino Nano, nRF24L01 module, and MPU6050 sensor. Proper wiring and component placement were done to minimize noise and ensure stable operation.

Drone Body

The drone body is designed to be sturdy, providing a stable platform for flight and camera operation. It houses all essential components, including the motors, propellers, ESCs (Electronic Speed Controllers), the axis control board, and the Arduino-based central controller. The frame supports an adaptive camera mounting system, allowing the camera to remain steady during flight for smooth video capture. The propeller arms are detachable, making the drone easier to transport and maintain. Power is supplied by a Li-Po battery, chosen for its high energy efficiency, ensuring longer flight times without adding unnecessary load. The design balances stability, portability, and functional efficiency, which makes it suitable for both learning purposes and basic aerial filming tasks.

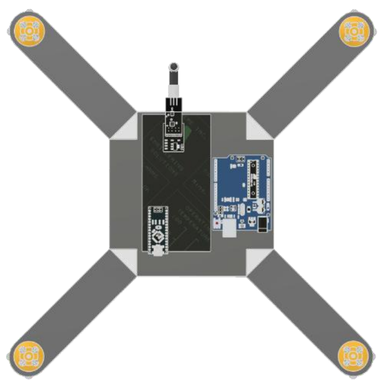

Figure 8: Top-View of Drone

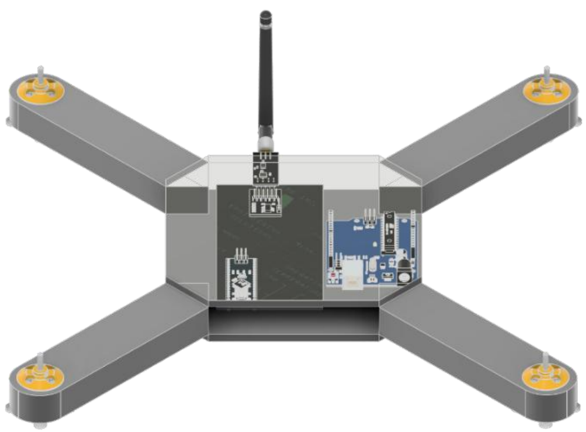

Figure 9: 3D View of Drone

Remote Controller

The remote controller is a 6-channel transmitter system designed using an Arduino Nano as the central control unit. It manages multiple input channels to control different drone movements such as throttle, rudder, elevator, and aileron, along with two auxiliary functions. Two joystick modules are used to generate analog signals corresponding to the drone's directional movements — throttle and rudder (CH3 & CH4), and aileron and elevator (CH1 & CH2).

The controller is powered by a 7.4V Li-Po battery, which supplies stable voltage through the Arduino's VIN pin. A switch and an LED indicator are included for power control and status indication. Additional components include a potentiometer (CH6 AUX2) and a toggle switch (CH5 AUX1), which allow for customizable control options, such as camera tilt or lighting.

or wireless communication, the setup includes an NRF24L01+PA+LNA module connected via a 3.3V adapter, enabling long-range, low-latency transmission to the drone's receiver. A 100–470 µF capacitor is used to stabilize the power supply to the RF module, ensuring consistent signal output.

Overall, this remote controller provides smooth analog control, reliable wireless transmission, and flexible auxiliary channels making it ideal for testing and controlling custom-built drones efficiently.

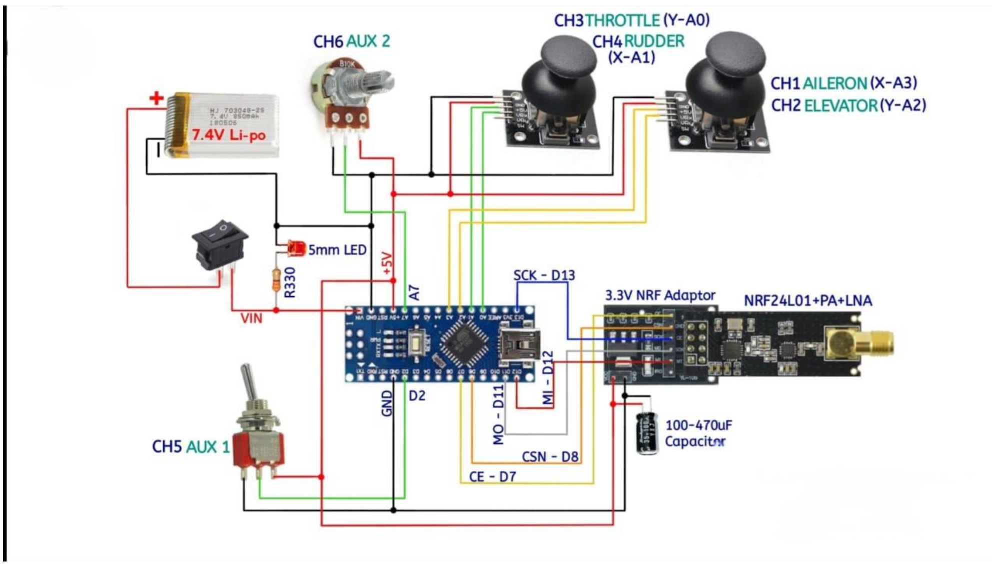

Figure 10: Transmitting Circuit

FUTURE SCOPE OF DRONE CAM

Future scope for Drone Cam technology looks extremely promising for cinematography, surveillance, and environmental monitoring. Coupled with more innovative video content, there was immense demand, and adaptive mounting systems in drones would play an even greater role in revolutionizing aerial cinematography, which can yield far higher variety and dynamism in camera angles from which filmmakers and content creators will benefit to unprecedented levels.

Battery technology is probably the field that most needs development. As it stands now, flight times for a drone are relatively minimal, although obviously enough, those will be changing with time as battery efficiency improves. And this will increase endurance for drones, meaning possibly longer filming periods without breaking and reassembling-the aspired film production. Professional shooting often requires long continuous shots.

Besides the improvement that is sure to be achieved in the life of the battery, cameras will benefit so much. The stabilizing systems of cameras enhance the quality of the shot and make it smooth even during high wind conditions and unlevel terrains. This gives a chance to have an environment to fly in; hence, suitable for both urban and remote areas.

Drones go beyond cinematography and apply other areas. For agriculture, it can monitor crops, detect diseases, and shortages in water provision. Drones can be useful for planting and observing the crops. For construction and surveying, drones would help observe buildings from an aerial view, collect information, and deliver real-time visuals about different aspects of project planning.

Further technological advance may result in the better and improved detection and avoidance of potential obstacles that could be involved with a drone, so this device could be safely used in a complex or cluttered setting and prove safer and more accessible to use. Future possibilities may even press further beyond AI-based controls, designed with an intent to make it easier to pilot, to allow time for creative rather than purely technological aspects.

It would gradually become more affordable and accessible to broaden its range of applications to cover an entire spectrum of industries, including cinematography and industrial inspections, to unlock the highest boundless possibilities.

CONCLUSION

Even though there is a lot of work left to do before Drone Cam technology becomes popular commercially, it shows great promise in aerial cinematography and surveillance.

An increasing number of researchers and companies are currently engaged in developing this concept, which offers solutions to the limitations of traditional static camera systems and restricted aerial views.

By using Drone Cam technology, industries can switch to more effective, flexible, and adaptable camera options, aiding in the progress of media production, environmental surveillance, and security uses.

Drone Cam's main idea provides a remedy for restrictions in traditional filming methods, enabling increased flexibility, accurate camera positions, and unique viewpoints that were previously impossible.

As a technology that is quickly advancing, Drone Cam is sparking advancements in aerial imaging and other related fields.

Future advancements in Drone Cam technology are predicted to expand its use in various fields such as cinematography and infrastructure inspection, showcasing its value.

This project successfully demonstrates the development of a quadcopter drone using low-cost components and a custom-built control system. It provided practical understanding of motor control, wireless communication, and sensor integration.

Although the system achieved basic functionality, further improvements are needed in stabilization algorithms and flight control precision.

REFERENCES

WEBSITES

[1] https://youtu.be/okxmLYSPywo?si=cCEHe_XpPJdHguu1

[2] https://youtu.be/GT0aYWnfdjc?si=cxk3YvZYWF9Qc-FM

[3] https://en.wikipedia.org/wiki/Drone

[4] https://cfdflowengineering.com/working-principle-and-components-of-drone/